

Arun Kumar Dara

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SUMMARY

Data Analyst with 3+ years of experience delivering real-time data solutions and predictive analytics across insurance, banking, and healthcare domains. Skilled in Python (Scikit-learn, XGBoost, LightGBM, CatBoost), SQL, R, PySpark, and cloud platforms including AWS, Azure, Google Cloud, and Databricks. Proven expertise in building scalable ETL pipelines, fraud detection models, anomaly detection, and regulatory-compliant data warehouses while developing dashboards in Power BI, Tableau, and Plotly Dash to drive actionable insights. Adept at cross-functional collaboration with actuarial, risk, underwriting, and operations teams, delivering measurable business impact such as improved fraud detection, reduced attrition, and accelerated reporting cycles.

TECHNICAL SKILLS

Languages:	Python, R, SQL (advanced CTEs, window functions, materialized views)
Libraries & Frameworks:	Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Statsmodels, Dask, ggplot2, UMAP, PCA
Visualization Tools:	Power BI, Tableau, Plotly Dash, R Shiny, Looker, ggplot2
Databases & ETL:	Snowflake, Delta Lake, BigQuery, SQL (CTEs, window functions, materialized views), Redis, ETL pipeline design
Cloud & Deployment:	AWS SageMaker, AWS Lake Formation, Azure Synapse, Google Cloud (BigQuery, Vertex AI, Pub/Sub, Dataflow, Data Catalog, DLP), Databricks, Apache Airflow, dbt
Big Data Technologies:	PySpark, Apache Beam, Databricks, Redis
Tools & Collaboration:	Neo4j, GraphQL APIs, Slack Webhooks, Notion, Confluence, JIRA, Excel
Methodologies:	Agile/Scrum, A/B Testing, Monte Carlo Simulation, Feature Engineering, Data Storytelling, Cross-functional Collaboration
Data Governance & Compliance:	Great Expectations, Collibra, AWS Lake Formation, Soda Core, Google Cloud DLP, SOX Compliance, FHIR/HL7 Standards

EXPERIENCES

Data Analyst | AIG, USA

Jun 2025 – Present

- Engineered real-time claims ingestion pipelines using Google Pub/Sub, Dataflow (Apache Beam), and BigQuery, securing sensitive FNOL (First Notice of Loss) data with Google Cloud DLP to meet regulatory compliance.
- Designed & implemented a claims data warehouse in BigQuery, leveraging advanced SQL transformations and Soda Core validations to achieve 99% data accuracy and accelerate actuarial reporting.
- Developed an anomaly detection model in Python (CatBoost) incorporating policy, exposure, and FNOL text features, improving fraud detection precision by 12% and reducing false positives by 23%.
- Automated & deployed anomaly scoring through Vertex AI Endpoints, integrating real-time alerts with Slack Webhooks, which reduced average fraud detection time by 35 minutes for SIU teams.
- Created & maintained interactive dashboards in Power BI, embedding row-level security and drill-downs to deliver tailored insights, cutting SIU triage cycles by 14%.
- Established governance practices by documenting lineage in Data Catalog and monitoring model drift with Evidently AI, ensuring sustainable model performance across business units.

Data Analyst Intern | AIG, USA

Jan 2025 – May 2025

- Assisted in designing a real-time claims ingestion pipeline using Google Pub/Sub, Dataflow, and BigQuery, ensuring accurate capture of FNOL (First Notice of Loss) data with Google Cloud DLP for sensitive information.
- Supported the development of a claims warehouse model in BigQuery, writing SQL transformations and performing Soda Core data quality checks that improved accuracy and reliability for reporting teams.
- Contributed to building an anomaly detection model in Python (CatBoost) by preparing training datasets, feature engineering, and evaluating performance metrics, which helped identify fraudulent claims more effectively.
- Collaborated on deployment of real-time scoring in Vertex AI and integration of fraud alerts via Slack Webhooks, reducing manual monitoring effort for claims investigators.
- Created interactive dashboards in Looker under mentorship, visualizing claims anomalies and trends, which supported SIU (Special Investigations Unit) teams in faster triage

Data Analyst | Hexaware technologies, India

Jan 2021 – July 2022

- Architected churn prediction models with Python (Scikit-learn, XGBoost) and AWS SageMaker, reducing customer attrition by 18% in the banking sector.
- Transformed healthcare and retail datasets using advanced SQL and Azure Synapse, accelerating actuarial risk assessments by 5x, securing \$2.5M new business, and boosting cross-sell revenue by 22%.
- Engineered healthcare data lakes with Delta Lake (FHIR/HL7) and centralized Snowflake marts, streamlining compliance audits and empowering faster data-driven underwriting.
- Built & visualized interactive analytics solutions by integrating manufacturing telemetry (PySpark, GraphQL APIs) and financial risk models (R Shiny, Monte Carlo simulations, ggplot2) into Power BI and Plotly Dash dashboards, enabling real-time defect tracking

and protecting ₹50+ crore AUM.

- Strengthened governance with Great Expectations and Collibra, automating financial data quality checks for SOX compliance and reducing reconciliation workloads.
- Enhanced credit scoring models by optimizing Databricks feature stores (UMAP, PCA), improving precision by 15%; additionally led workshops on reusable analytics frameworks via Notion/Confluence, doubling deployment speed of data science solutions.

Data Analyst | Mphasis, India

Aug 2022 – July 2023

- Engineered scalable data pipelines using SQL, Apache Airflow, and dbt to integrate cross-border banking transactions into Snowflake, enhancing AML anomaly detection by 30% and reducing compliance risks.
- Developed & optimized predictive models for customer lifetime value (CLV) using Python (LightGBM, SHAP) on Databricks by consolidating structured & semi-structured CRM/transactional data, driving higher upsell conversions and insurance policy renewals.
- Standardized healthcare claims data by designing normalization engines with PySpark and FHIR/HL7 frameworks, shortening patient risk scoring cycles from 2 weeks to 3 days and enabling precision-medicine analytics.
- Synthesized & visualized multimodal logistics datasets via GraphQL APIs and Neo4j, building Tableau dashboards that enabled inventory rebalancing and cut warehouse stockouts.
- Instituted robust data quality frameworks with Great Expectations and AWS Lake Formation, automating validation checks for sensitive banking data and reducing manual exception handling by 40%.
- Designed & evaluated scenario-based A/B experiments with R (ggplot2, Statsmodels), presenting insights through interactive dashboards that optimized digital loan origination funnels and increased conversions.
- Accelerated advanced analytics initiatives by streamlining feature engineering for high-frequency trading signals using Python (Pandas, Dask) and Redis, reducing training cycles by 50%; additionally, championed data storytelling practices via Notion/Confluence, doubling adoption of self-service analytics across business units.

EDUCATION & CERTIFICATIONS

Masters in Science in Management – Data Analytics Specialization

May 2025

Indiana Wesleyan University, USA

Bachelor of Technology in Engineering

Aug 2022

TKR College of Engineering & Technology, India

- **Power BI Data Analyst Associate**

PROJECTS

Patient Condition Classification Using Drug Reviews

- Designed an end-to-end NLP pipeline using Python (Pandas, NumPy, and Scikit-learn) to process 10,000+ patient drug reviews, extracting key side effects and sentiment indicators.
- Built and optimized classification models (Logistic Regression, Random Forest, XGBoost) with GridSearchCV/RandomizedSearchCV, achieving 85% accuracy in predicting patient conditions.
- Automated preprocessing workflows (tokenization, lemmatization, TF-IDF, sentiment scoring), cutting manual data cleaning by 40% and enabling scalable clinical insights.
- Integrated demographic and health data to enrich predictions and deployed REST APIs via Flask for real-time drug effectiveness checks by healthcare practitioners.
- Delivered actionable insights through interactive dashboards (Power BI, Tableau) and structured reports, improving treatment personalization and clinical decision-making.

Stock Market Analysis for Reliance Industries

- Analyzed 8 years of stock data via Python & Yahoo Finance API, uncovering historical trends to guide trading strategies.
- Developed ARIMA, LSTM, and Random Forest forecasting models with technical indicators (MACD, RSI, and Bollinger Bands), achieving robust price predictions.
- Automated data pipelines for continuous updates and integrated real-time dashboards in Streamlit, enabling investors to act on live forecasts.
- Designed portfolio optimization and quantitative trading algorithms, balancing returns with risk in volatile markets.
- Delivered detailed risk & performance reports, helping stakeholders evaluate scenarios and make data-driven investment decisions.

Book Recommendation System with Chatbot Integration

- Built a hybrid recommendation engine combining collaborative filtering and NLP-based content models to deliver personalized book suggestions.
- Integrated a conversational AI chatbot for seamless discovery, with A/B tested dialogue flows that boosted user satisfaction.
- Deployed the solution via Flask & Streamlit with a PostgreSQL-backed preference database for adaptive learning of user tastes.
- Leveraged external APIs for real-time metadata enrichment and presented engagement insights through interactive dashboards.
- Enhanced UX by analyzing behavior metrics and implementing improvements, increasing user retention by 30%.